

# Cristian Garces

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## Personal Profile

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**Cristian** is a **PhD Candidate** in the Department of Computer Science at **Tulane University**. He will receive his Ph.D. degree in May 2027. His research applies machine learning to binary analysis and reverse engineering, focusing on recovering function names and other semantic information from stripped binaries and on evaluating whether that information helps practitioners with vulnerability analysis. A second thread is cybersecurity education research: he is developing auto-gradable reverse-engineering exercises generated from real-world CVEs, closing the gap between the synthetic binaries students typically practice on and the real programs they will encounter in practice (currently under review at CSET '26). He has also led hands-on binary-analysis workshops for high school students. Cristian has taught in sixteen course assignments across two institutions. He is best prepared to teach theory of computation, which he has supported for five semesters, and the computer systems sequence (computer systems, computer organization, and information security), for which he has served as instructor of record for the computer systems lab, supervising two undergraduate course assistants, and as teaching assistant for the lecture courses; he has also served as a teaching assistant for discrete structures and object-oriented programming and as a tutor for operating systems. He has received a departmental Outstanding Graduate Teaching Award at both institutions and, as chapter president of Tau Beta Pi, organized midterm review sessions that reached roughly 160 undergraduates.

## Education

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- Ph.D. Tulane University**, Computer Science – New Orleans, LA USA 2020 – 2027 (expected)
- Advisor: Dr. Jiang Ming
  - Transferred with advisor from **University of Texas at Arlington** (2020 – Fall 2022)
- B.Sc. University of Texas at Arlington**, Computer Science and Engineering, *summa (Hons) cum laude* – Arlington, TX USA 2016 – 2020
- Honors Thesis: *SCREWIT: A COMPUTER VISION APPROACH TO IDENTIFY SCREWS AND FASTENERS*; advised by Dr. Christopher McMurrough
  - Undergraduate Research Assistant, Department of Physics, UTA HEP (2018 – 2020); advised by Dr. Jaehoon Yu

## Research Interests

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- AI-Assisted Binary Analysis
- Binary Reverse Engineering
- Vulnerability Analysis
- Cybersecurity Education

## Teaching

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### Tulane University

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|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| CMPS 3300 – Software Studio, Teaching Assistant                                                                                                        | Fall 2026   |
| CMPS 3510 – Computer Organization, Teaching Assistant                                                                                                  | Fall 2026   |
| <b>CMPS 2301 – (2 sections) – Intro. to Computer Systems and Networking Lab, Instructor of Record</b> ; supervised two undergraduate course assistants | Spring 2026 |
| CMPS 2300 – Intro. to Computer Systems and Networking, Teaching Assistant                                                                              | Spring 2026 |
| CMPS 4770 – Operating Systems, Tutor                                                                                                                   | Spring 2025 |
| CMPS 4410/6410 – Information Security, Teaching Assistant (Spring 2023, Fall 2023, Spring 2024, Fall 2024)                                             | 2023 – 2024 |

## University of Texas at Arlington

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|----------------------------------------------------------------------------------------------------|-------------|
| CSE 3315 – Theoretical Concepts, Teaching Assistant (Spring, Summer, Fall 2021; Summer, Fall 2022) | 2021 – 2022 |
| CSE 2315 – Discrete Structures, Teaching Assistant                                                 | Spring 2022 |
| CSE 1325 – Object-Oriented Programming in C++, Teaching Assistant                                  | Fall 2020   |

## Awards

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| <b>Outstanding Graduate Teaching Award</b> , Department of Computer Science, Tulane University                                  | 2023 |
| <b>Outstanding Graduate Teaching Award</b> , Department of Computer Science and Engineering, University of Texas at Arlington   | 2022 |
| <b>Lockheed Martin Missiles and Fire Control Graduate Fellowship</b> , College of Engineering, University of Texas at Arlington |      |

## Conference Publications

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- [1] **[APSEC '26] Cristian Garces**, Keith Mitchell, Kun Liu, Chenke Luo, Haotian Zhang, Jiang Ming. Reassessing Function-Name Recovery in Stripped Binaries: Preprocessing, Metrics, and Reproducibility. In *Proceedings of the 33rd Asia-Pacific Software Engineering Conference*, 2026. To appear.
- [2] **[ISPEC '26] Cristian Garces**, Keith Mitchell, Chenke Luo, Haotian Zhang, Jiang Ming. Do Recovered Names Help? Evaluating Binary Function-Name Recovery for Vulnerability Triage In *Proceedings of the International Conference on Information Security Practice and Experience*, 2026. To appear.

## Papers under Review

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- [1] **Cristian Garces**, Matthew Toups, Jiang Ming. LINEUP: Generating Auto-Gradable Reverse-Engineering Exercises from Real Vulnerable Binaries. Under review at **CSET '26** (Workshop on Cyber Security Experimentation and Test, co-located with ACSAC 2026).
- [2] Binlin Cheng, Chenke Luo, **Cristian Garces**, Xiaolei Ren, Yan Lin, Jiang Ming. Thwarting Exact DNN Decompilation: Harnessing Compiler-Induced Diversity to Protect Neural Executables Under review at **NDSS '27**.

## Professional Service

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| <b>External Reviewer</b> , USENIX Security Symposium | 2024, 2025 |
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## Outreach Activities

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| <b>Chapter President</b> , Tau Beta Pi, University of Texas at Arlington                                                                                                                                                    | 2021 – 2022 |
| <b>Organizer</b> , undergraduate CS midterm review sessions, Tau Beta Pi, University of Texas at Arlington (four sessions across two semesters, ~40 students each)                                                          | 2021 – 2022 |
| <b>Teaching Assistant</b> , OurCS@DFW (VIRUS: How to Take Apart A Computer Virus), K-12 outreach workshop on introductory binary analysis, Department of Computer Science and Engineering, University of Texas at Arlington | Spring 2022 |
| <b>Judge</b> , FIRST Lego League, FIRST Robotics                                                                                                                                                                            | 2024        |

## Professional Memberships

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|----------------------------------------------|------------|
| Tau Beta Pi (Engineering Honor Society)      | 2019 – Now |
| Upsilon Pi Epsilon (Computing Honor Society) | 2022 – Now |